

CAP477:PROGRAMMING IN JAVA

L:3 T:0 P:2 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: familiar with basic concepts of Java programming to solve the real-world problems

CO2 :: understand the Object-oriented programming principles to write efficient and reusable codes and lambda expressions

CO3 :: evaluate exception handling and input/output techniques to improve the robustness and reliability of Java applications

CO4 :: construct collections and generics to ensure clean, robust, and maintainable Java code

Unit I

Introduction to Java : history and features of Java, understanding JDK, JRE and JVM, Java program structure, writing simple Java class and main() method, command-line arguments, identifiers, keywords and variables, using primitive data types, type conversion, access modifiers, static keyword, wrapper class, arithmetic operator, assignment operator, relational operator, logical operator, working with bit-wise operator, unary and ternary operator, operator precedence, using if/else constructs and switch-case statements

Unit II

Loops, array and strings : working with for loop, while loop, do-while loop and for-each loop, fundamentals about arrays, array access and iterations, multi-dimensional arrays, using varargs, enumerations, basics of class and object, writing constructors and methods, this keyword, initializer blocks, overloading methods and constructors, constructors and methods of String and String Builder class

Unit III

Inheritance and Polymorphism : inheritance, using super and final keywords, method overriding, overriding toString() method, overriding equals() method, instanceof operator, Object class, abstract method and abstract class, interfaces, static and default methods, using swing components to demonstrate inheritance

Unit IV

Functional interface, lambda expressions and utility class : using lambda expressions, implementing threads, implementing listener using lambda expressions, understanding the importance of static and non-static nested classes, local and anonymous class, working with Dates using utility class

Unit V

Exceptions, assertions and I/O fundamentals : exception overview, exception class hierarchy and exception types, using try catch and finally for exception handling, usage of throw and throws, handling multiple exceptions using multi-catch, creating custom exceptions, propagation of exceptions, auto-close resources with try-with resources statement, testing invariants by using assertions, describing the basics of input and output in Java, read and write data from the console, using streams to read and write files, writing and read objects using serialization

Unit VI

Collections and Generics : creating a collection by using generics, implementing an ArrayList, implementing TreeSet using Comparable and Comparator interfaces, implementing a HashMap, implementing a Deque, creating a custom generic class, using the type inference diamond to create an object, using bounded types and wild cards

List of Practicals / Experiments:

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- program to implement a java class
- program to demonstrate the instantiation of class and accessing the attributes using object of class
- program to demonstrate the uses of String and StringBuilder
- program to implement polymorphism and using proper access control
- program to demonstrate the use of abstract class and nested class

- program to demonstrate the inheritance and its importance using Swing Components
- program to implement multithreading using Lambda Expressions
- program to demonstrate the use of all the keywords used for exception handling and need of assertion
- program to implement read and write operation using console and File
- program to demonstrate the use of list of items

Text Books:

1. PROGRAMMING WITH JAVA by E.BALAGURUSAMY, MC GRAW HILL

References:

1. JAVA 2: THE COMPLETE REFERENCE, FIFTH EDITION by HERBERT SCHILDT, MC GRAW HILL
2. INTRO TO JAVA PROGRAMMING, COMPREHENSIVE VERSION by Y. DANIEL LIANG, PEARSON